TCP SYN Port Scan — Security Assessment Report

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1. Executive Summary

This report documents a controlled TCP SYN port scan conducted against a Windows target host within an isolated virtual laboratory, together with the corresponding detection and analysis performed from the defensive side. The objective was to reproduce the reconnaissance phase of a typical network attack and to establish the observable characteristics by which such activity can be identified in network traffic.

A SYN scan was executed from a Linux host against the target using Nmap. The resulting traffic was captured on the target with Wireshark and analysed. The scan was positively identified through a distinct traffic pattern: a single source host issuing a high number of half-open TCP connection attempts across a wide range of destination ports, with no completed three-way handshakes. The activity corresponds to MITRE ATT&CK technique T1046 (Network Service Discovery).

The exercise confirmed that port scanning, while non-destructive in itself, is reliably detectable through both network packet analysis and host-based logging, and that early detection of reconnaissance provides valuable warning ahead of follow-on attack activity.

2. Objective

The objectives of this assessment were:

- To perform a TCP SYN port scan against a controlled target host.
- To capture and analyse the associated network traffic from the defender's perspective.
- To document the observable indicators that distinguish port scanning from legitimate network activity.
- To define appropriate detection and mitigation measures.

3. Scope and Test Environment

All activity was confined to an isolated VMware host-only network with no connectivity to production systems or the internet. The scope was limited to the two hosts listed below.

Attacker host: Linux virtual machine, IP address 192.168.213.132 Target host: Windows virtual machine, IP address 192.168.213.133 Network segment: 192.168.213.0/24 (isolated, host-only) Monitoring: Wireshark packet capture running on the target host

No systems outside this segment were in scope, and no production or third-party assets were affected at any point.

4. Tools Used

Nmap was used on the attacker host to perform the port scan. Wireshark was used on the target host to capture and analyse the resulting network traffic. Standard Windows command-line utilities (ipconfig, netstat) were used to support host-side verification.

5. Methodology

The assessment followed a straightforward reconnaissance-and-detection methodology. Network connectivity between the two hosts was first confirmed. A packet capture was then started on the target host to ensure all scan traffic would be recorded. The port scan was executed from the attacker host, after which the capture was stopped and the traffic analysed to identify the scan signature and correlate it with host-based indicators.

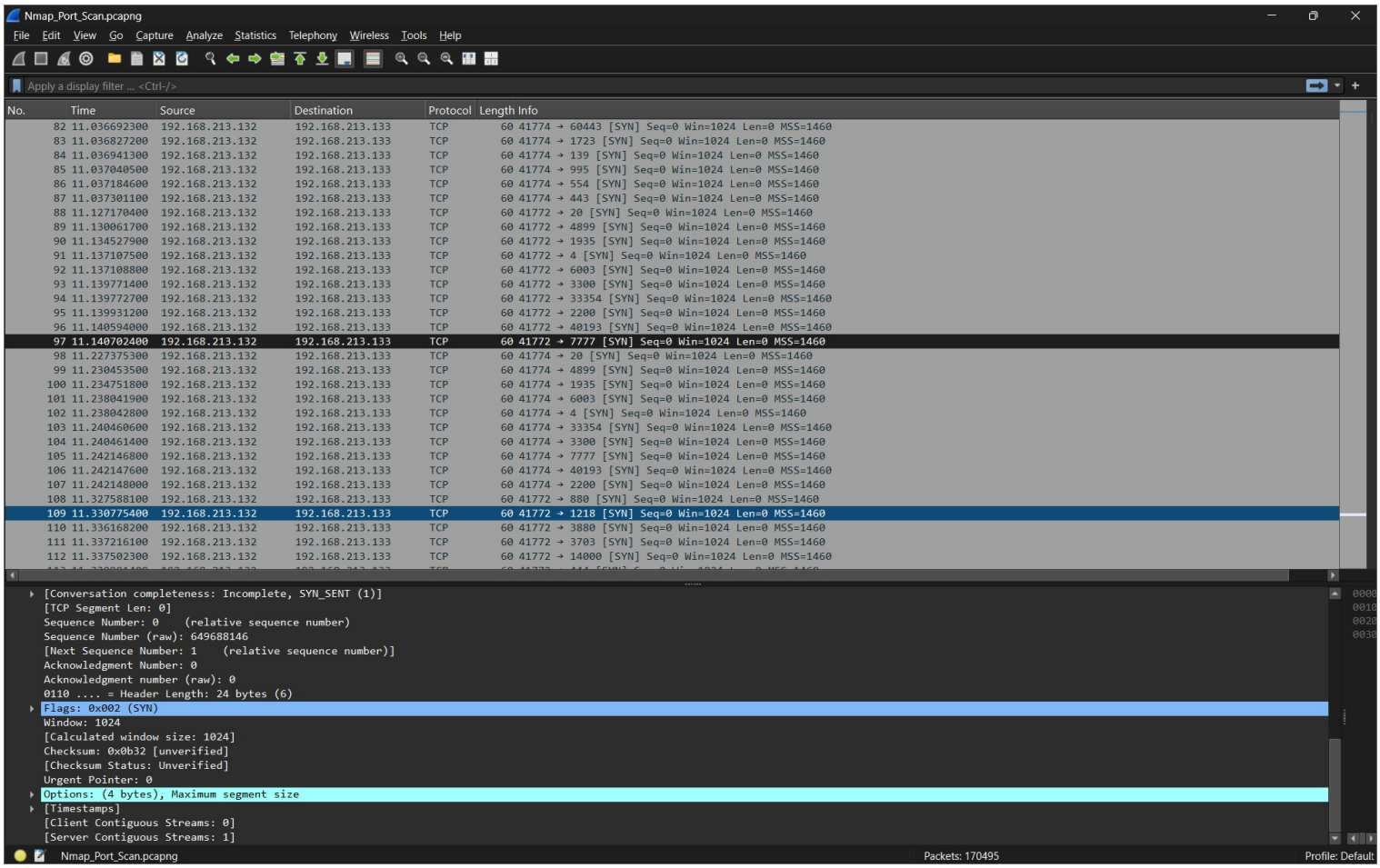
6. Execution

The scan was performed using a TCP SYN (half-open) scan:

```
sudo nmap -sS 192.168.213.133
```

In a SYN scan, Nmap transmits a TCP packet with only the SYN flag set to each target port and evaluates the response. A SYN/ACK response indicates an open port, a RST response indicates a closed port, and the absence of any response indicates a filtered port. The scanning host does not complete the handshake with a final ACK; the connection is therefore left half-open. This behaviour is the primary basis for later detection. By default the scan evaluated the most common TCP ports and completed within a few seconds.

7. Observations and Evidence



Analysis of the captured traffic produced the following findings.

All observed packets originated from the single source address 192.168.213.132 and were directed to the single target address 192.168.213.133. This one-to-one relationship, in which a single host contacts a single target across numerous ports, is not characteristic of normal client activity.

The destination port values varied continuously across successive packets, covering a broad range including ports 139, 443, 554, 995, 1723, 1935, 4899, 6003, 7777, 33354, 40193 and 60443, among others. The ports were not contacted in sequential order, consistent with Nmap's randomisation of scan order.

Every captured packet carried only the SYN flag. Inspection of individual packets confirmed the TCP flags field as 0x002 (SYN). The packets were uniform in size and structure, each 60 bytes in length with a window size of 1024, a maximum segment size of 1460, and no payload. The source port alternated between a small number of values (41772 and 41774) across the scan.

The TCP analysis field for the captured packets reported a conversation state of "Incomplete, SYN_SENT", confirming that no three-way handshake was completed for any probed port. Several destination ports appeared more than once in the capture, consistent with Nmap retransmitting probes to ports that did not respond to the initial attempt.

The following Wireshark display filter was applied to isolate the relevant traffic:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

This filter retained only packets with the SYN flag set and the ACK flag clear. The filtered output showed a single source address generating connection attempts across a large number of destination ports within a very short interval. The Statistics and Conversations view corroborated this, showing a single host with numerous short, incomplete TCP conversations.

8. Detection and Analysis

The defining signature of the activity is a single source host generating a high volume of lone SYN packets to many distinct destination ports with no completed handshakes, within a short time window. This pattern is distinguishable from legitimate traffic by the breadth of destination ports and the consistent absence of handshake completion.

Beyond packet capture, the activity can be detected through host-based telemetry. With dropped-packet logging enabled, the Windows Firewall log records repeated blocked connection attempts from a single source across multiple ports. The netstat utility shows transient SYN_RECEIVED entries during the scan, and Resource Monitor reflects a short burst of inbound connection activity from one remote address. In an enterprise environment, a SIEM platform such as Wazuh can be configured to alert when a single source generates a high number of firewall drops to distinct ports within a defined interval. It should be noted that a SYN scan generates limited native Windows event logging, so network-layer detection or firewall logging is generally required for reliable identification.

9. Impact Assessment

A port scan is non-destructive and does not by itself compromise the target. Its significance lies in its role as a precursor activity: it enables an attacker to enumerate exposed services and identify potential entry points prior to an intrusion attempt. On a default Windows configuration,

commonly exposed ports such as 135, 139 and 445 (RPC and SMB services) represent likely targets for subsequent exploitation. The risk associated with the scan itself is therefore assessed as low, but its presence should be treated as an early indicator of potential targeting.

10. Recommendations

The following measures are recommended to reduce exposure to, and improve detection of, port scanning activity:

- Disable or close all network services that are not required, reducing the number of exposed ports.
- Configure the host firewall to drop, rather than reject, unsolicited inbound SYN packets, so that the scanning host cannot reliably distinguish open from closed ports.
- Deploy a network intrusion detection system, such as Suricata, to automatically alert on scan behaviour.
- Implement SIEM correlation rules to flag reconnaissance patterns across host and network logs.

11. Conclusion

The TCP SYN port scan was successfully executed and positively identified through network traffic analysis. The activity was characterised by a single source host issuing numerous half-open connection attempts across a wide range of destination ports, with no completed handshakes. The exercise demonstrated that reconnaissance activity of this type is reliably detectable through both packet analysis and host-based logging, and reinforced the value of identifying such activity early in the attack lifecycle.

This assessment was conducted in an isolated laboratory environment on systems owned by the author, for educational and defensive-research purposes only.